

REAL-TIME VOICE · 17 MIN READ

Real-Time Conversational AI Voice Agents & Telephony Systems

Building Sub-500ms Voice Bots with WebRTC, LiveKit, Silero VAD, and Twilio

Executive Architecture Summary: The complete technical blueprint for architecting ultra-low-latency conversational voice agents. Covers WebRTC audio streaming, Voice Activity Detection (Silero VAD), turn-taking orchestration, Twilio SIP trunking, and sub-500ms response pipelines.

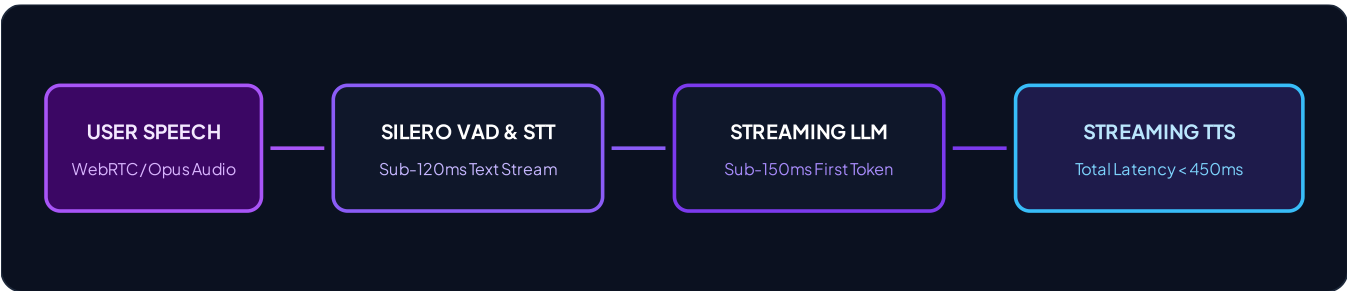
1. The Anatomy of Real-Time Voice Latency

Human conversation begins to feel unnatural and awkward when latency exceeds 600ms. Traditional serial pipelines (record entire utterance -> STT -> LLM response -> TTS -> playback) take 2.5 to 4 seconds, completely breaking conversational flow. Achieving fluid, human-like voice agents requires streaming audio frames at every stage.

- Streaming Speech-to-Text: Deepgram Nova-2 streaming WebSocket transcription returns word tokens in 120ms.
- Streaming Token Generation: Fast LLMs (Gemini Flash, Groq Llama-3, GPT-4o-mini) emit first tokens within 150ms.
- Streaming Text-to-Speech: Cartesia Sonic or ElevenLabs Flash synthesizes audio chunks within 100ms.
- End-to-End Budget: Sub-500ms total latency delivers instantaneous, seamless conversational rhythm.

2. The Voice Agent Streaming Architecture

Audio frames stream continuously over WebRTC / WebSocket channels through a centralized coordinator that manages barge-in and interruptions.



3. Voice Activity Detection (VAD) & Barge-In Handling

Handling interruptions naturally is the hardest engineering challenge in conversational voice. When a user speaks while the agent is talking, the system must immediately kill audio playback and clear the LLM generation buffer.

```
// Real-Time Barge-In & Interruption Handler (Node / LiveKit)
class VoiceSessionCoordinator {
  constructor(audioOutputTrack, llmStream) {
    this.outputTrack = audioOutputTrack;
    this.llmStream = llmStream;
    this.isAgentSpeaking = false;
  }

  onUserSpeechDetected() {
    if (this.isAgentSpeaking) {
      console.log('[BARGE-IN] User interrupted agent! Aborting audio playback...');
      // 1. Immediately mute and flush audio playback queue
      this.outputTrack.clearBuffer();
      // 2. Cancel inflight LLM generation stream
      this.llmStream.abort();
      // 3. Reset agent speaking state
      this.isAgentSpeaking = false;
    }
  }
}
```

- Silero VAD: ML-based voice activity detection distinguishes human speech from background clicks in 30ms.
- Adaptive Noise Cancellation: Filter out room echo to prevent the agent from interrupting itself.

4. Telephony Integration & Twilio SIP Trunking

Bridge WebRTC voice agents to standard PSTN telephone networks using Twilio Media Streams or LiveKit SIP connectors. Automate inbound customer support and outbound appointment confirmation.

- Dual-Track Audio: Separate incoming caller audio from outbound agent synthesis to enable clean acoustic separation.
- Call Transfer Protocol: Seamless warm handoff to human operators when the caller requests human escalation.